Evaluating Large Language Models Trained on Code (Codex)

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A language model for code

Program synthesis with language models

- Program synthesis is a longstanding challenge (Simon, 1963; Manna et al., 1971).
- Large code corpora became available, e.g. **CodeSearchNet** and **The Pile**.
- Self-supervised objectives used in NLP were adapted to code: **CodeBERT**, **T5**, and **PyMT5** are representative examples.
- Early analysis suggested that **GPT-3** could generate programs from Python docstrings even without explicit code-specialized training.
- Hypothesis of the paper: specializing a GPT language model for code could work across many tasks.

This work: Codex

Codex is a GPT-style model specialized for code. The paper positions it as a model that can generate useful code across tasks and notes that it is used for **Copilot**.

Refs: Chen et al. (2021); Husain et al. (2019); Gao et al. (2020); Feng et al. (2020); Brown et al. (2020); Wang et al. (2021).

Related Work

Learning to Execute

LSTMs are trained on programs and can learn algorithmic tasks such as adding two 9-digit numbers with about 99% accuracy. Curriculum learning plays a key role.

Naturalness of software

This work studies code with n-gram language models and shows that source code is statistically natural in ways that can be exploited for modeling and tooling.

SPoC

The SPoC dataset contains about 18K C++ programs and studies pseudocode-to-code synthesis using a seq2seq model with best-first search.

APPS

APPS introduces a benchmark with 10K programming problems across three difficulty levels and includes test cases for evaluating generated solutions.

Refs: Zaremba et al. (2014); Hindle et al. (2012); Kulal et al. (2019); Hendrycks et al. (2021).

Task and approach

Task: functions from docstrings

- Input: **docstring**
- Output: **Python function**
- Code correctness is evaluated automatically via **unit tests**.
- This differs from natural language generation, which usually needs human assessors or heuristic metrics.
- For benchmarking, the paper constructs **164 original programming problems** with tests.
- Problem types include language comprehension, algorithms, simple mathematics, and interview-style questions.

Problem solving with one sample

Approach: generate samples and check whether any pass the unit tests.

- Codex (12B): **28.8%**
- Codex (300M): **13.2%**
- GPT-J (6B): **11.4%**
- Other GPT models: approximately **0%**
- After supervised fine-tuning on correct implementations, Codex-S (12B) reaches **37.7%**

Problem solving with multiple samples

Programming usually involves iteration and bug fixing. Repeated sampling approximates this process.

- Codex-S (12B) with 100 samples solves **77.5%** of problems.
- Selecting the sample with highest mean log-probability solves **44.5%**.

Refs: Chen et al. (2021); Wang et al. (2021).

Evaluation framework

Functional correctness

- Match-based metrics compare against a reference exactly or fuzzily (e.g. BLEU).
- Such metrics are limited for code because many different programs can be equivalent.
- The paper therefore emphasizes **functional correctness**: a sample is correct if it passes a set of unit tests.
- This is also how human developers judge code in practice, e.g. via test-driven development and CI.

The `pass@k` metric

Generate k samples per problem. A problem is solved if *any* sample passes the tests.

Unbiased estimator used in the paper:

$$\texttt{pass@}k \triangleq \mathbb{E}_{\text{problems}}\left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}\right]$$

where $n \geq k$ is the number of samples and c is the number of correct samples among them.
Advantage: using more generated samples reduces variance while keeping comparisons fair.

Refs: Chen et al. (2021); Papineni et al. (2002); Ren et al. (2020); Roziere et al. (2020); Kulal et al. (2019).

Evaluation details

HumanEval: hand-written evaluation set

- Functional correctness is evaluated on **164 hand-written problems**.
- Each HumanEval problem contains a function signature, a docstring, a body, and several unit tests (average: **7.7** tests per problem).
- Hand-written problems matter because the models are trained on GitHub, where many public benchmark solutions already exist.
- HumanEval targets simple mathematics, reasoning, and language comprehension.
- The dataset is made public for future benchmarking.

Sandbox for executing generated programs

- Executing public code and generated code creates a security risk.
- The paper therefore runs untrusted programs in a sandboxed environment.
- **gVisor** is used to isolate the host, and network-adjacent services are protected with eBPF-based firewall rules.

Refs: Chen et al. (2021); Hendrycks et al. (2021); OpenAI HumanEval repository; gVisor.

Nuances of pass@k estimation

Estimating `pass@k`

Aim: estimate the probability that among k samples, at least one is correct.

- If the true probability of a correct sample is p, then:

$$\Pr(\text{none correct}) = (1-p)^k, \qquad \mathrm{pass@}k = 1-(1-p)^k.$$

- A naive plug-in estimate $1-(1-\hat{p})^k$ from $\hat{p} = \mathrm{pass@}1$ systematically underestimates performance.
- It also makes results look better simply by drawing more samples from the same pool.
- The paper's estimator instead compares tasks across different numbers of samples in a consistent way.

Figure placeholder

Insert the two estimator-comparison plots from the original slide here.

Top: $1-(1-\hat{p})^k$ estimator
Bottom: unbiased combinatorial estimator

Key point

Slightly higher initial variance is acceptable because the estimator enables fairer comparisons across different sample budgets.

Refs: Chen et al. (2021).

Nuances of pass@k estimation

Why the proposed estimator is unbiased

The second term in

$$\mathbb{E}\left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}\right]$$

directly estimates the probability of drawing **only failed samples** when choosing k items without replacement from n candidates.

Setup

- Let $c \sim \mathrm{Binom}(n, p)$, where p is $\mathrm{pass@1}$.
- If $n - c < k$, then $\binom{n-c}{k} = 0$, so the estimator returns 1.
- Otherwise, the estimator computes the probability that all k chosen samples are failures.

Sketch of the derivation

$$\mathbb{E}\left[\frac{\binom{n-c}{k}}{\binom{n}{k}}\right] = \sum_{i=0}^{n-k} \frac{\binom{n-i}{k}}{\binom{n}{k}} \binom{n}{i} p^i (1-p)^{n-i}$$

$$= (1-p)^k \sum_{i=0}^{n-k} \binom{n-k}{i} p^i (1-p)^{n-k-i} = (1-p)^k$$

Therefore,

$$\mathbb{E}[\mathrm{estimator}] = 1 - (1-p)^k = \mathrm{pass@}k.$$

Refs: Chen et al. (2021).

Implementing the pass@k estimator

Numerically stable implementation

```
def pass_at_k(n, c, k):
    """
    n: total number of samples
    c: number of correct samples
    k: k in pass@k
    """
    if n - c < k:
        return 1.0
    return 1.0 - np.prod(
        1.0 - k / np.arange(n - c + 1, n + 1)
    )
```

Rationale for $n - c < k$

If fewer than k failed samples exist, then it is impossible to draw k failures without replacement:

$$\binom{n-c}{k} = 0 \quad \Rightarrow \quad 1 - \frac{\binom{n-c}{k}}{\binom{n}{k}} = 1.$$

Rationale for $n - c \geq k$

Rewrite the combinatorial ratio as a stable product:

$$\frac{\binom{n-c}{k}}{\binom{n}{k}} = \prod_{j=n-c+1}^{n} \left(1 - \frac{k}{j}\right)$$

which is exactly what the NumPy implementation computes.

Refs: Chen et al. (2021).

Code Fine-tuning

Overview

Codex is produced by fine-tuning GPT models of up to **12B** parameters. Unlike GPT models trained only on text, Codex obtains non-trivial HumanEval performance, and with 100 samples per problem, the majority of problems have at least one solution.

Data collection and fine-tuning

- Training corpus collected in May 2020 from **54M GitHub repositories**.
- Produced **179 GB** of unique Python files under 1 MB.
- Filtering removed likely auto-generated files, very long lines, and files with a very small fraction of alphanumerics.
- Final result: **159 GB** of unique Python files.
- Surprisingly, fine-tuning from GPT-3 gave no improvement over training from scratch, although it converged faster and was used in practice.

Optimization and tokenization

- Same learning rate as the corresponding GPT model.
- Linear warmup for 175 steps; cosine decay afterwards.
- Training on **100B tokens** with Adam ($\beta_1 = 0.9,\ \beta_2 = 0.95,\ \epsilon = 10^{-8}$) and weight decay 0.1.
- GPT-3 tokenization is reused, but extra whitespace tokens are added for code.
- This reduces the token count for code by roughly **30%**.

Refs: Chen et al. (2021); Kingma and Ba (2015).

Supervised Fine-tuning

Overview

GitHub data contains far more than clean “function from docstring” examples: it also includes classes, config files, scripts, and data files. This mismatch may limit HumanEval performance, so the paper builds a supervised training set of standalone function-completion tasks.

Source 1: competitive programming problems

- Problems, solutions, and function signatures were collected from interview-preparation and programming-contest websites.
- Problem descriptions are used as docstrings.
- Unit tests are created from examples in statements and by submitting incorrect solutions to reveal hidden cases.
- These sources provide self-contained tasks with strong algorithmic content and usually good test coverage.
- Roughly **10,000** problems are curated from these sources.

Refs: Chen et al. (2021).

Supervised Fine-tuning

Source 2: problems from Continuous Integration

- Problems are also extracted from open-source repositories by tracing inputs and outputs during integration tests with `sys.setprofile`.
- CI config files reveal how to create environments, install dependencies, and run tests.
- Repositories using **Travis** or **Tox** are particularly useful; additional source is pulled from PyPI.
- Because the code is untrusted, all integration tests are executed in the sandbox.
- Only about **40,000** problems are collected from millions of functions, mainly because not all functions are easy to trace and some runtime objects cannot be restored outside the sandbox.

Filtering problems

- Automatically gathered prompts may be ambiguous or stateful.
- Codex-12B is used to generate 100 samples per problem.
- If all samples fail, the problem is discarded as too hard or ambiguous.
- Verification is rerun multiple times to filter out stateful problems.

Refs: Chen et al. (2021); Travis CI; Tox.

Supervised Fine-tuning

Methodology - training with prompts

Training examples are assembled in the same prompt format used at evaluation time.

```
def solution(lst):
    """Given a non-empty list of integers, return the sum
    of all odd elements that are in even positions.

    Examples
    solution([5, 8, 7, 1]) => 12
    solution([3, 3, 3, 3, 3]) => 9
    solution([30, 13, 24, 321]) => 0
    """
    return sum(lst[i] for i in range(0, len(lst))
               if i % 2 == 0 and lst[i] % 2 == 1)
```

- Codex is fine-tuned on these training problems to produce **Codex-S**.
- Training minimizes the negative log-likelihood of the reference solution while masking the prompt.
- Shorter prompts are left-padded so the target solutions line up.
- Learning rate is 1/10 of Codex with the same schedule until validation loss plateaus (after fewer than 10B tokens).

Optimal temperatures

- Codex-S prefers higher temperatures than Codex for all cases with $k > 1$.
- The paper suggests this may reflect a narrower learned distribution.
- For later evaluation:
  - $T^* = 0$ for `pass@1`
  - $T^* = 1$ for `pass@100`

Insert the “best temperature for different k” plot here
(Codex vs. Codex-S)

Refs: Chen et al. (2021).

Comparative Analysis of Related Models

Related approaches

- GPT-Neo and GPT-J-6B are the closest comparison points.
- Both are trained on The Pile, of which about 8% comes from GitHub.
- GPT-J-6B already produces qualitatively reasonable code.

Temperatures

GPT-Neo: 0.2, 0.4, 0.8
GPT-J-6B: 0.2, 0.8
Tabnine: 0.4, 0.8

HumanEval results

Model	$k = 1$	$k = 10$	$k = 100$
GPT-Neo 125M	0.75%	1.88%	2.97%
GPT-Neo 1.3B	4.79%	7.47%	16.30%
GPT-Neo 2.7B	6.41%	11.27%	21.37%
GPT-J 6B	11.62%	15.74%	27.74%
Tabnine	2.58%	4.35%	7.59%
Codex-12M	2.00%	3.62%	8.58%
Codex-25M	3.21%	7.10%	12.89%
Codex-42M	5.06%	8.80%	15.55%
Codex-85M	8.22%	12.81%	22.40%
Codex-300M	13.17%	20.37%	36.27%
Codex-679M	16.22%	25.70%	40.95%
Codex-2.5B	21.36%	35.42%	59.50%
Codex-12B	28.81%	46.81%	72.31%

Takeaway

Codex-12B goes considerably beyond the performance of prior models.

Parameter note

The slide emphasizes that Codex-12B achieves this despite having roughly **20x fewer parameters than GPT-J-6B**.

Refs: Chen et al. (2021); Black et al. (2021); Wang et al. (2021); Gao et al. (2020).

Results on the APPS Dataset

APPS dataset comparison

- APPS was introduced to measure coding challenge competence.
- It contains **5000 train** and **5000 test** coding problems.
- Each example includes unit tests; training examples also include solutions.
- Most APPS tasks are **full-program synthesis** problems (stdin/stdout), not single-function generation.
- Original APPS metrics: **strict accuracy** and **test case average**.
- Codex results here are reported only under strict accuracy / `pass@k`.

Implementation details and results

Additional factors on the slide:

- Example cases: APPS gives 3 I/O examples; Codex uses one example as a formatting hint (“1-shot”).
- Timeouts: results are reported both with and without counting solutions that pass tests but time out after 3 seconds.

Method	Intro	Interview	Competition
GPT-Neo 2.7B raw `p@1`	3.90	0.57	0.00
GPT-Neo 2.7B raw `p@5`	5.50	0.80	0.00
1-shot Codex raw `p@1`	4.14 (4.33)	0.14 (0.30)	0.02 (0.03)
1-shot Codex raw `p@5`	9.65 (10.05)	0.51 (1.02)	0.09 (0.16)
1-shot Codex raw `p@100`	20.20 (21.57)	2.04 (3.99)	1.05 (1.73)
1-shot Codex raw `p@1000`	25.02 (27.77)	3.70 (7.94)	2.33 (5.85)
1-shot Codex filtered `p@1`	22.78 (25.10)	2.64 (5.78)	3.04 (5.25)
1-shot Codex filtered `p@5`	24.52 (27.15)	3.23 (7.13)	3.08 (5.53)

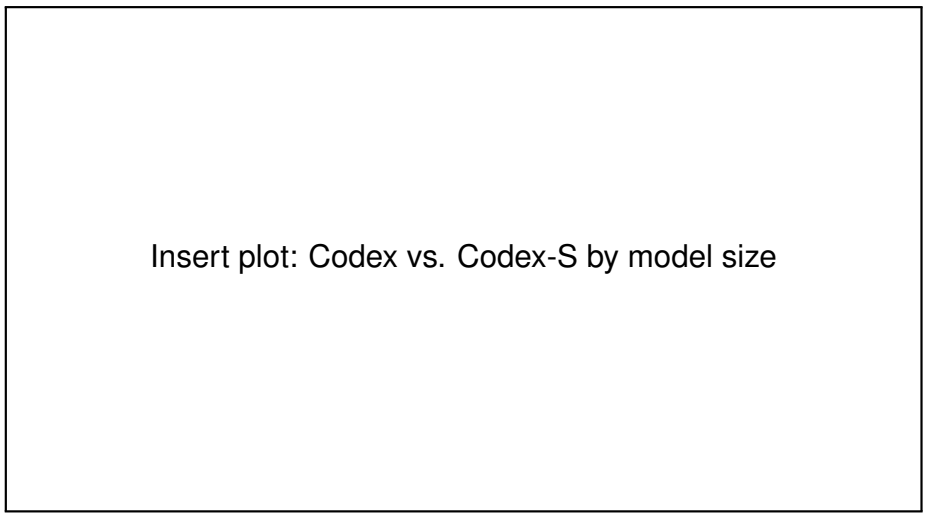
Temperature 0.6 is used for all k in `pass@k`.

Refs: Chen et al. (2021); Hendrycks et al. (2021); Black et al. (2021).

Supervised Fine-tuning: Results

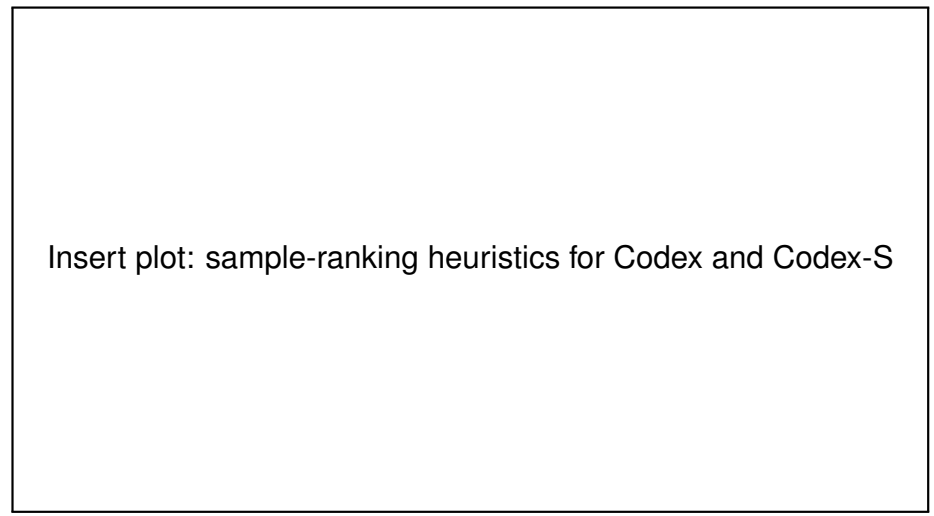
Influence of model size

- Codex-S beats Codex by an average of **6.5%** on `pass@1`.
- On `pass@100`, the average gain is **15.1%**.
- The slide shows both curves increasing with model size.



Sample ranking heuristics

- Oracle reranking remains an upper bound.
- Mean log-probability is again the best practical heuristic.
- The average benefit of mean log-probability is about **2% higher** for Codex-S than for Codex.



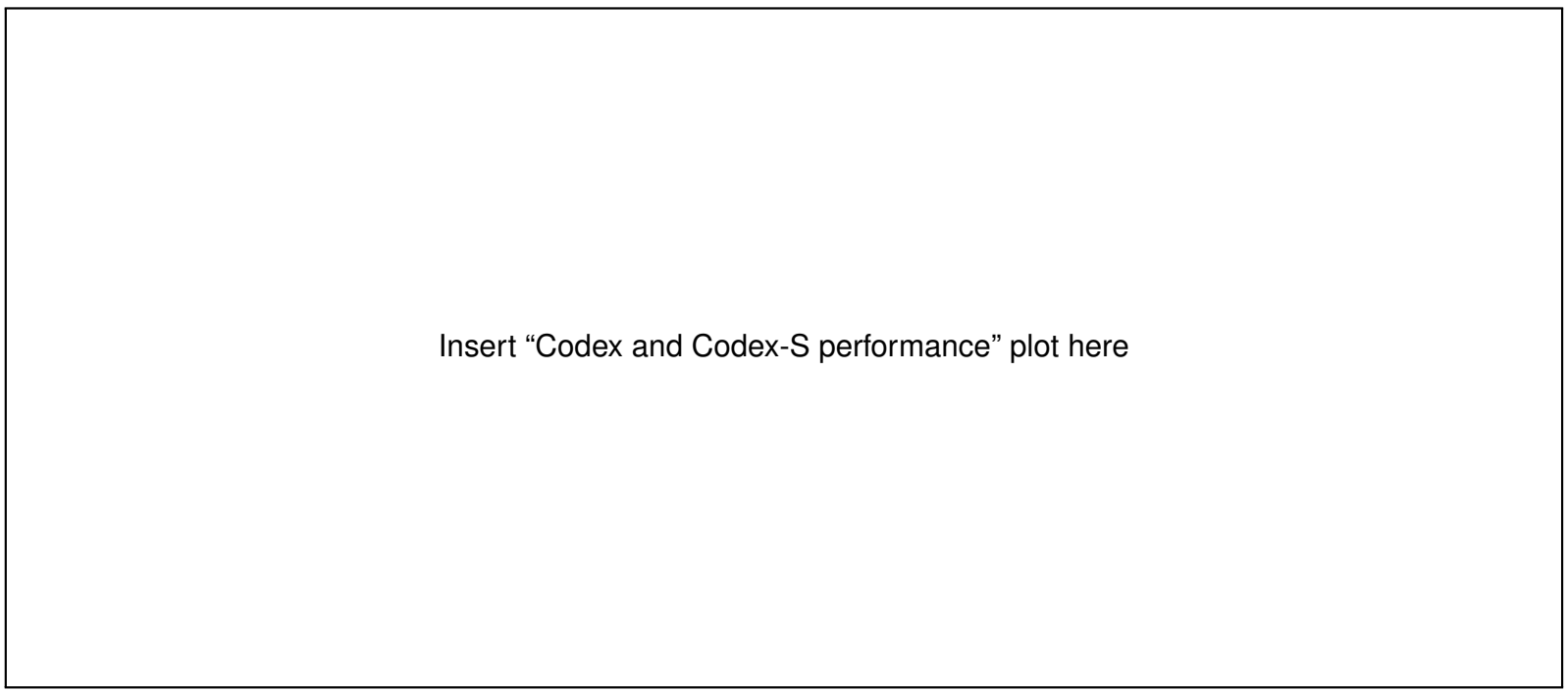
Refs: Chen et al. (2021).

Comparing Codex and Codex-S

Comparing training strategies on HumanEval

The original slide shows a single scaling plot across model sizes for GPT-3, Codex, and Codex-S, together with reranking variants.

- **Codex-S mean-logp reranking** solves **44.5%** when reranking 100 samples.
- **Codex-S oracle reranking** solves **77.5%** when selecting from 100 samples using the unit tests.



Refs: Chen et al. (2021).

Docstring complexity

Composing building blocks

The 13 building blocks are chained by concatenating:

- their one-line descriptions into a docstring, and
- their one-line implementations into a code body.

Example from the slide:

```
def string_manipulation(s: str):
    """
    This function takes a string as input, then
    returns the result of performing the following
    sequence of manipulations:
    - make every other character uppercase
    - replace spaces with triple spaces
    """
    s = "".join(char.upper() if i % 2 == 0 else char
                for i, char in enumerate(s))
    s = s.replace(" ", "   ")
    return s
```

Results

Insert “Synthetic pass rate vs. components” plot here

- As each component is added, the pass rate drops by about **2–3x**.
- Human programmers do not show the same degradation once they can chain two operations.
- Codex also makes variable-binding mistakes when many variables are mentioned.

Example binding failure

Prompt idea: “Add 3 to `y`, then subtract 4 from both `x` and `w`. Return the product of the four numbers.”
Observed failure: the generated code subtracts 4 from `x` but forgets to subtract 4 from `w`, and only multiplies two numbers.

Refs: Chen et al. (2021).

Broader Impacts and Hazard Analysis

Applications of Codex

Potentially useful applications listed on the slide:

- onboarding users to new codebases,
- reducing context switches for experienced coders,
- enabling non-programmers to write specifications,
- producing draft implementations,
- aiding education and exploratory coding.

Hazard analysis and over-reliance

- Main risks: generated code may not match user intent, and the system may be misused.
- The analysis is framed in terms of risk factors rather than full product safety design.
- Over-reliance is a central concern, especially for novice programmers.
- Generated code may look correct but still be wrong or insecure.
- Automation bias makes this especially dangerous: people often trust automated suggestions too much.
- Human oversight remains necessary.

Refs: Chen et al. (2021); Leveson (2019).

Summary

Summary

- The paper studies whether large language models can be trained to generate code from docstrings.
- After GitHub fine-tuning, Codex performs well on human-written problems, especially easy interview-style tasks.
- Performance improves further when training is brought closer to the evaluation distribution and when multiple samples are considered.
- Codex-D is introduced for code-to-docstring generation; it is weaker than code generation but still comparable.
- The paper also emphasizes limitations, broader impacts, and safety risks such as over-reliance, bias, misalignment, insecurity, and economic change.

Refs: Chen et al. (2021).

Backup / Q&A

Prompting for evaluation

Prompting to compute `pass@k`

Each HumanEval problem is assembled into a prompt made from the function signature, the docstring, and any examples.

```
def incr_list(l: list):
    """Return list with elements incremented by 1.
    >>> incr_list([1, 2, 3])
    [2, 3, 4]
    >>> incr_list([5, 3, 5, 2, 3, 3, 9, 0, 123])
    [6, 4, 6, 3, 4, 4, 10, 1, 124]
    """
```

Sampling continues until one of the following stop tokens is encountered: `'\nclass'`, `'\ndef'`, `'\n#'`, `'\nif'`, or `'\nprint'`.

Sampling details

- Nucleus sampling with $p = 0.95$ is used for all sampling evaluation.
- For the simple single-function prompt on the slide, Codex 12B reaches about **pass@1 = 0.9**.

Multi-function prompts

The paper notes a dramatic degradation when the prompt contains multiple functions. In the example on the slide, Codex 12B gets only about **pass@1 = 0.005**.

Refs: Chen et al. (2021); Holtzman et al. (2019).

Loss scaling and temperature

Loss scaling

- Language model losses often follow a power law.
- The paper finds a similar pattern for Codex test loss on a held-out GitHub validation set.
- Approximate scaling law shown on the slide:

$$\left(\frac{N}{5.92 \times 10^7}\right)^{-0.13}$$

- Takeaway: Codex fine-tuning appears to scale smoothly with model size.

Insert loss-scaling plot here

Sampling temperature

- Best temperature depends on k.
- For larger k, higher temperatures work better because `pass@k` rewards whether the model generates *any* correct solution.
- The slide shows both `pass@k` vs. k for several temperatures and a curve for the best temperature as a function of k.

Insert temperature-vs-`pass@k` plots here

Refs: Kaplan et al. (2020); Chen et al. (2021).

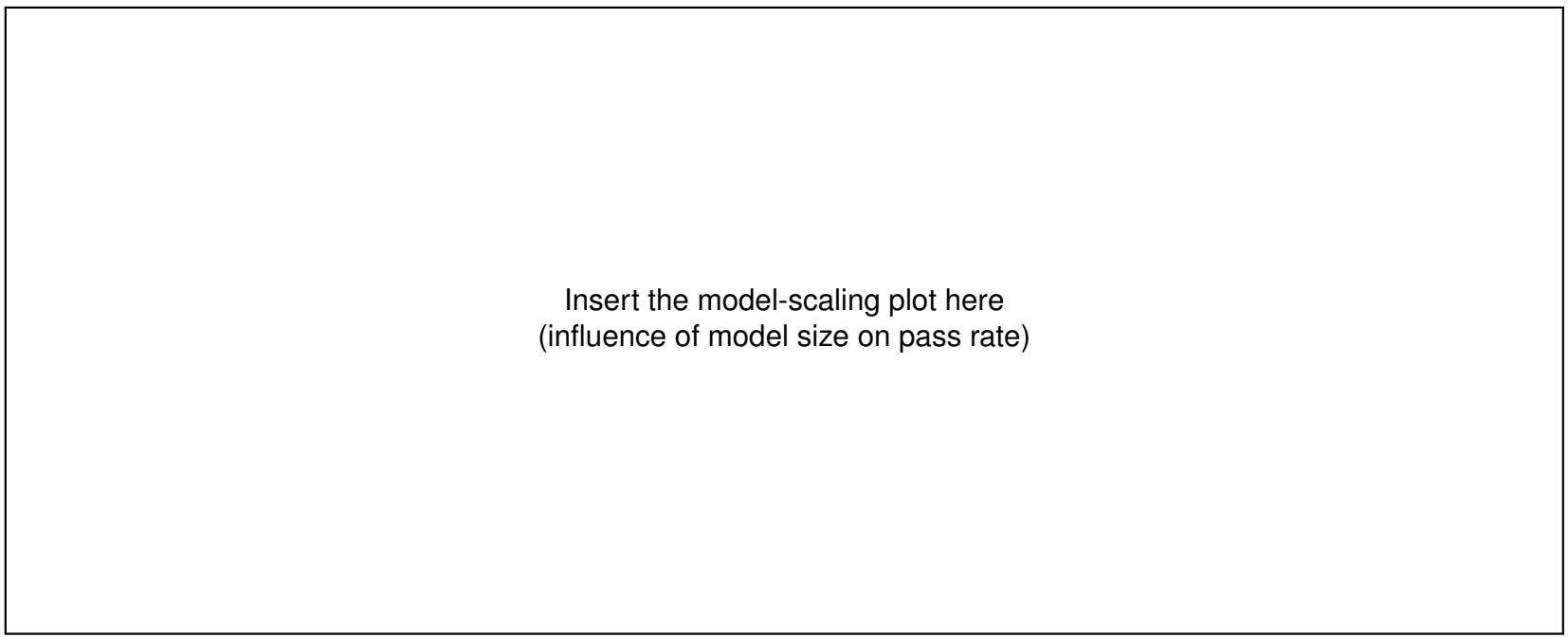
Model scaling at optimal temperatures

Model scaling

For a **679M**-parameter model, the best temperatures reported on the slide are:

- $T^* = 0.2$ for `pass@1`
- $T^* = 0.8$ for `pass@100`

The main takeaway is that performance scales smoothly as a function of model size.

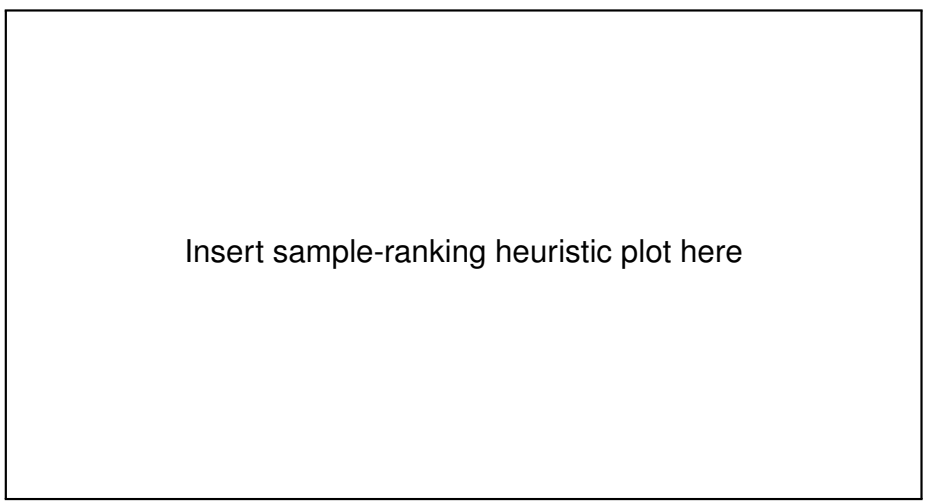


Refs: Chen et al. (2021).

Sampling heuristics and BLEU score

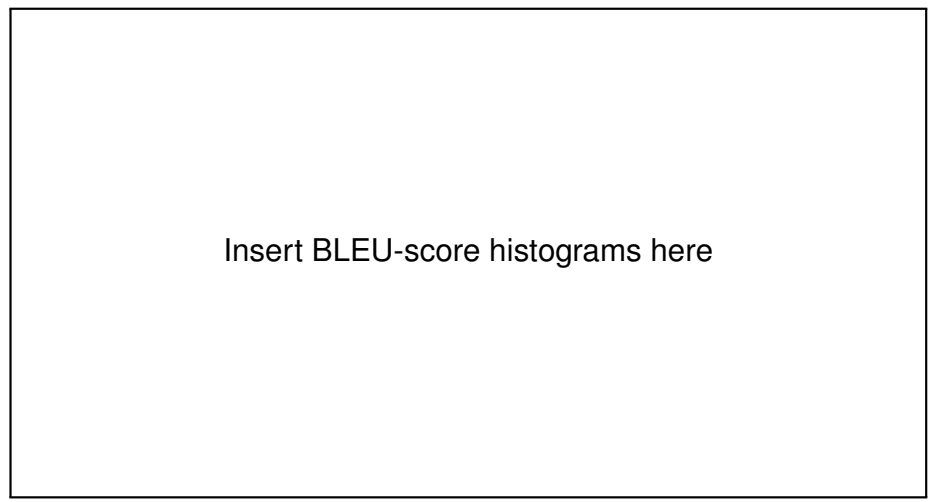
Effectiveness of sampling heuristics

- `pass@k` can be interpreted as evaluating the best out of k samples.
- An oracle that knows the unit tests is the upper bound.
- Without an oracle, a practical goal is to select one sample among k candidates.
- For Codex 12B at $T = 0.8$, the best non-oracle heuristic is **mean log-probability**.



BLEU score correlation

- BLEU scores are computed for HumanEval Codex 12B samples at $T = 0.8$.
- The comparison is made against reference solutions.
- Correct and wrong solutions are **not cleanly separable** by BLEU.



Refs: Chen et al. (2021).

Code generation examples

Prompt

```
def is_prime(n):
    """Return true if a given number is prime.
    >>> is_prime(6)
    False
    >>> is_prime(101)
    True
    >>> is_prime(4)
    False
    >>> is_prime(1)
    False
    """
```

8 random samples are shown on the original slide (temperature 0.8).

Completions 1–3 (wrong)

- **Completion 1:**

```
for i in range(2, n):
    if n % i == 0:
        return False
return True
```

  Fails for $n = 1$ because it returns True.

- **Completion 2:** exactly the same as Completion 1.
- **Completion 3:**

```
# TODO: implement this function
pass
```

  Fails all unit tests.

Completions 4–5

- **Completion 4 (correct):** handles $n < 2$, handles 2, rejects evens, and tests odd divisors up to $\sqrt{n}$.
- **Completion 5 (wrong):** uses a helper to test divisibility up to $\sqrt{n}$, but still returns True for $n = 1$.

Refs: Chen et al. (2021).

Code generation examples

Completions 6–8

- **Completion 6 (correct):**

```
prime = True
if n == 1:
    return False
for i in range(2, n):
    if n % i == 0:
        prime = False
return prime
```

  Exhaustive check, but correct.

- **Completion 7 (wrong):** repeats the same bug as Completion 1.
- **Completion 8 (correct):** efficient primality test based on checking divisibility by 2, 3, and then numbers of the form $6k \pm 1$.

Why $6k \pm 1$ matters

For primes greater than 3, any candidate factor must be of the form $6k \pm 1$.

- Numbers of the form $6k$, $6k + 2$, and $6k + 4$ are divisible by 2.
- Numbers of the form $6k + 3$ are divisible by 3.
- So after handling 2 and 3, only $6k - 1$ and $6k + 1$ remain worth testing.

This makes Completion 8 faster than checking every number up to n.

Refs: Chen et al. (2021); standard primality-test observation.